



2026 IEEE INTERNATIONAL CONFERENCE ON
ACOUSTICS, SPEECH, AND SIGNAL PROCESSING (ICASSP 2026)

May 04 - 08, 2026

Barcelona, Spain



When Differential Privacy Meets Wireless Federated Learning: An Improved Analysis for Privacy and Convergence

Yaoling Chen¹, Hao Liang^{1,2}, Xiaotong Tu¹

¹School of Informatics, Xiamen University, China

²Internet of Things Thrust,

The Hong Kong University of Science and Technology (Guangzhou), China

MLSP-L31.4: Optimization and Privacy in Federated Learning

Fri, 8 May, 14:00 - 16:00

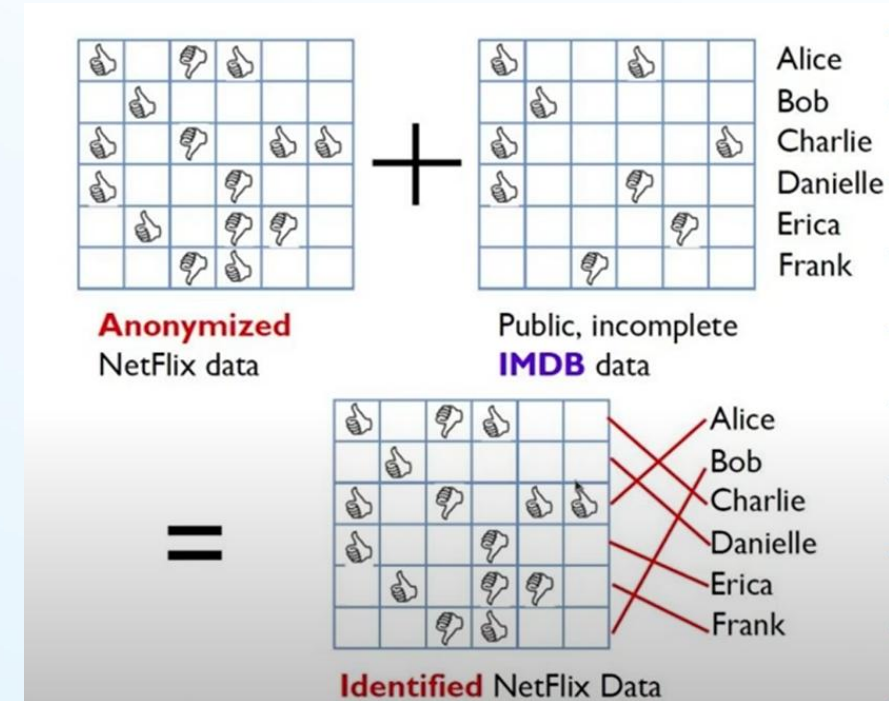
2026.ieeeicassp.org

Contents

- I. Introduction
- II. Related Works and Motivations
- III. The Proposed Method
- IV. Numerical Results

I . Why Privacy Protection Matters

- **Anonymization is not enough for privacy!**
 - Netflix Prize Competition
 - Contestants were given a dataset of user ratings where identifying information was anonymized and competed to create a recommendation engine
 - Researchers were able to de-anonymize the dataset by linking user ratings to public IMDB ratings
 - Memorization in Neural Networks [1]
 - My social security number is xxx-xx-xxxx
 - Memorize? Find Secrets?
- **How to address such privacy risks?**



[alina class lecture](#)
[Lecture 1A: Some Attempts at Data Privacy - NYC Taxis and Netflix](#)

I . Federated Learning

- A promising distributed learning procedure
 - Federated learning (FL) [1]
 - enabling multiple devices to collaboratively train a shared model **without exposing their local data**
 - significant advantages in terms of data privacy and communication efficiency
 - Wireless federated learning (WFL)
 - over-the-air FL (AirFL) [2], achieves scalable and communication-efficient training
- However, naive privacy preservation by FL is **not sufficient against emerging privacy attacks**, such as membership inference attacks [3]
- Ideally, we need a way to learn general trends of the dataset **without revealing any individual' s private information/contribution**

[1] McMahan B, Moore E, Ramage D, et al. Communication-efficient learning of deep networks from decentralized data[C]//Artificial intelligence and statistics. PMLR, 2017: 1273-1282.

[2] Yang K, Jiang T, Shi Y, et al. Federated learning via over-the-air computation[J]. IEEE transactions on wireless communications, 2020, 19(3): 2022-2035.

[3] Nasr M, Shokri R, Houmansadr A. Comprehensive privacy analysis of deep learning: Passive and active white-box inference attacks against centralized and federated learning[C]//2019 IEEE symposium on security and privacy (SP). IEEE, 2019: 739-753.

I . Differential Privacy

- Differential privacy (DP) for machine learning (ML)
 - A formal performance metric with theoretical guarantee [1], and widely used in protect sensitive data in training of ML models [2]-[4]
 - the key idea is to inject carefully **calibrated randomness**, typically in the form of additive noise, to mask the contribution of any data sample
 - **investigate wireless federated learning with differential privacy**

$$\underbrace{\ln \left(\frac{\mathbb{P}[A(D') \in O]}{\mathbb{P}[A(D) \in O]} \right)}_{\text{"Privacy loss"}} \leq \underbrace{\epsilon}_{\text{Bound on privacy loss, the "privacy budget"}}$$

Question 1: What is the privacy loss of differentially private WFL (DPWFL) as a function of the number of iterations?

- [1] Dwork C, Roth A. The algorithmic foundations of differential privacy[J]. Foundations and trends® in theoretical computer science, 2014, 9(3–4): 211-407.
- [2] Abadi M, Chu A, Goodfellow I, et al. Deep learning with differential privacy[C]//Proceedings of the 2016 ACM SIGSAC conference on computer and communications security. 2016: 308-318.
- [3] Kim M, Günlü O, Schaefer R F. Federated learning with local differential privacy: Trade-offs between privacy, utility, and communication[C]//ICASSP 2021-2021 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP). IEEE, 2021: 2650-2654.
- [4] McMahan H B, Ramage D, Talwar K, et al. Learning differentially private recurrent language models[J]. arXiv preprint arXiv:1710.06963, 2017.

Contents

- I. Introduction
- II. Related Works and Motivations
- III. The Proposed Method
- IV. Numerical Results

II. Related Work and Motivations

- Prior arts for privacy analysis [1]-[4]:
 - based on composition theorems [5] or moment accountant [6], [7]
 - both of which assume that **all intermediate models are revealed during training** and thus tend to over-estimate privacy loss
 - **Overly conservative noise injection**, degraded model utility, and introduces unnecessary communication overhead

Question 2: Can we more accurately track the privacy trajectory throughout training?

[1] Liu D, Simeone O. Privacy for free: Wireless federated learning via uncoded transmission with adaptive power control[J]. IEEE Journal on Selected Areas in Communications, 2020, 39(1): 170-185.

[2] Hasircioğlu B, Gündüz D. Private wireless federated learning with anonymous over-the-air computation[C]//ICASSP 2021-2021 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP). IEEE, 2021: 5195-5199.

[3] Park S, Choi W. On the differential privacy in federated learning based on over-the-air computation[J]. IEEE Transactions on Wireless Communications, 2023, 23(5): 4269-4283.

[4] Liu H, Yan J, Zhang Y J A. Differentially private over-the-air federated learning over MIMO fading channels[J]. IEEE Transactions on Wireless Communications, 2024, 23(8): 8232-8247.

[5] Dwork C, Roth A. The algorithmic foundations of differential privacy[J]. Foundations and trends® in theoretical computer science, 2014, 9(3-4): 211-487.

[6] Mironov I. Rényi differential privacy[C]//2017 IEEE 30th computer security foundations symposium (CSF). IEEE, 2017: 263-275.

[7] Abadi M, Chu A, Goodfellow I, et al. Deep learning with differential privacy[C]//Proceedings of the 2016 ACM SIGSAC conference on computer and communications security. 2016: 308-318.

II. Related Work and Motivations

- Prior arts for privacy analysis [1]-[3]:
 - based on privacy amplification by iteration framework [1]
 - strengthen privacy guarantees by avoiding the release of intermediate results
 - Often lead to privacy bounds that **increase ad infinitum with iterations** or require **restrictive (strongly) convexity assumptions**
- Recent progress [4] shows that privacy loss can converge under general smooth objectives and established a tight convergence guarantee in the centralized setting
 - its implications for wireless federated learning **remain unexplored**

[1] Feldman V, Mironov I, Talwar K, et al. Privacy amplification by iteration[C]//2018 IEEE 59th Annual Symposium on Foundations of Computer Science (FOCS). IEEE, 2018: 521-532.

[2] Chourasia R, Ye J, Shokri R. Differential privacy dynamics of Langevin diffusion and noisy gradient descent[J]. Advances in Neural Information Processing Systems, 2021, 34: 14771-14781.

[3] Altschuler J, Talwar K. Privacy of noisy stochastic gradient descent: More iterations without more privacy loss[J]. Advances in Neural Information Processing Systems, 2022, 35: 3788-3800.

[4] Liang H, Zhang W, He X, et al. An improved privacy and utility analysis of differentially private SGD with bounded domain and smooth losses[C]//Proceedings of the AAAI Conference on Artificial Intelligence. 2026, 40(28): 23401-23408.

II. Related Work and Motivations

- Prior arts for utility (convergence) analysis:
 - ignore the critical **role of gradient clipping** (an essential mechanism for enforcing DP)
 - hinge on **restrictive assumptions** (e.g., convexity or bounded sample-wise gradient), which are often violated in practice, especially in training deep neural networks

Question 3: How can we establish convergence guarantees for general non-convex objectives while explicitly accounting for gradient clipping?

-
- [1] Liu D, Simeone O. Privacy for free: Wireless federated learning via uncoded transmission with adaptive power control[J]. IEEE Journal on Selected Areas in Communications, 2020, 39(1): 170-185.
- [2] Hasircioğlu B, Gündüz D. Private wireless federated learning with anonymous over-the-air computation[C]//ICASSP 2021-2021 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP). IEEE, 2021: 5195-5199.
- [3] Park S, Choi W. On the differential privacy in federated learning based on over-the-air computation[J]. IEEE Transactions on Wireless Communications, 2023, 23(5): 4269-4283.
- [4] Liu H, Yan J, Zhang Y J A. Differentially private over-the-air federated learning over MIMO fading channels[J]. IEEE Transactions on Wireless Communications, 2024, 23(8): 8232-8247.

II. Contributions

- To address the above questions, we provide an improved analysis for the privacy and utility of DPWFL while incorporating both device selection and mini-batch sampling:
 - we derive a precise privacy characterization for DPWFL by jointly incorporating device-level and data-level sampling, and prove that it **can converge to a constant**
 - we rigorously **characterize the effect of gradient clipping** on DPWFL, and establish an **improved privacy-utility trade-off** for general non-convex smooth objective
 - we experimentally validate our theoretical findings

Contents

- I. Introduction
- II. Related Works and Motivations
- III. The Proposed Method
- IV. Numerical Results

III. DPWFL Algorithm

Objective: $f^* = \min_{\theta} f(\theta) \triangleq \frac{1}{n} \sum_{i=1}^n f_i(\theta)$

1. On Client:

- Receive global model from server
- Local update with mini-batch stochastic gradients and gradient clipping
- Transmit scaled clipped gradients over the wireless channel

2. On Server:

- Receive superimposed signals via multi-access channel with channel noise
- Apply combining to obtain an estimate of the aggregated gradient
- Global update
- Broadcast updated model to all clients

Algorithm 1: DPWFL

```

1 Input: learning rate  $\eta$ , number of communication
   rounds  $T$ , number of local rounds  $Q$ , active device
   ratio  $p$ , mini-batch sampling ratio  $q$ , and clipping
   threshold  $c$ 
2 Initialize  $\theta_i^{(0)} = \theta^{(0)}$ 
3 while  $t < T$  do
4   On devices  $i \in \mathcal{I}^{(t)}$  (in parallel):
5      $\theta_i^{(t,0)} \leftarrow \theta^{(t)}$ ;
6     Local update with gradient clipping:
        $\sum_{\xi \in \mathcal{B}_i^{(t)}} \text{clip}_c(\nabla l(\theta_i^{(t)}; \xi))$  with a mini-batch
        $\mathcal{B}_i^{(t)}$  and  $\text{clip}_c(x) = x \min(1, c/\|x\|)$ ;
7     Transmit  $\mathbf{x}_i^{(t)} = s_i^{(t)} \sum_{\xi \in \mathcal{B}_i^{(t)}} \text{clip}_c(\nabla l(\theta_i^{(t)}; \xi))$ ;
8   end
9   On server:
10    Receive  $\mathbf{y}^{(t)} = \sum_{i \in \mathcal{I}^{(t)}} h_i^{(t)} \mathbf{x}_i^{(t)} + \zeta^{(t)}$  with
       channel noise  $\zeta^{(t)}$ ;
11    Combining:  $\hat{\nabla} f(\theta^{(t)}) = (\gamma^{(t)})^{-1} \mathbf{y}^{(t)}$ ;
12    Global update:  $\theta^{(t+1)} = \theta^{(t)} - \frac{\eta}{pqn} \hat{\nabla} f(\theta^{(t)})$ ;
13    Broadcast  $\theta^{(t+1)}$  to all  $n$  devices;
14  end
15   $t \leftarrow t + 1$ ;
16 end
17 Output:  $\theta^{(T)}$ 

```

III. Assumptions for Analysis

- We outline the assumptions for our analysis, which are commonly used in non-convex optimization and FL analysis

Assumption 1 (*L*-smooth) *The loss function $l(\cdot; \xi)$ is L -smooth, i.e., for all θ, θ' , we have $\|\nabla l(\theta; \xi) - \nabla l(\theta'; \xi)\| \leq L \|\theta - \theta'\|$.*

Assumption 2 (Bounded Parameter Domain) *For any iteration $t \in \{0, \dots, T-1\}$, the domain of the model parameters has diameter $D > 0$.*

Assumption 3 (Bounded Variance) *For any $i \in [n]$ and $\theta \in \mathbb{R}^d$, the gradient of loss function, $\nabla l(\theta; \xi)$, has bounded variance, i.e., $\mathbb{E} [\|\nabla l(\theta; \xi) - \nabla f_i(\theta)\|^2] \leq \sigma_l^2$.*

Assumption 4 (Gradient Dissimilarity) *For any $i \in [n]$ and $\theta \in \mathbb{R}^d$, the gradient of local empirical loss function, $\nabla f_i(\theta)$, satisfies $\mathbb{E} [\|\nabla f_i(\theta) - \nabla f(\theta)\|^2] \leq \sigma_g^2$.*

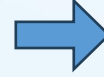
III. Privacy Analysis for DPWFL

1. Scaled Channel Noise Decoupled

$$(\gamma^{(t)})^{-1} \zeta^{(t)}$$

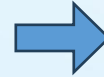
Wireless Noisy Mechanism

together with the server aggregation procedure



Remaining Noise

reduce the shift amount of the shifted Rényi divergence



$$\sqrt{1 - \beta^{(t)}} (\gamma^{(t)})^{-1} \zeta^{(t)}$$

2. Privacy Evaluation of Wireless Noisy Mechanism

Lemma 3.1 (Wireless Noisy Reduction) Define

$$\psi^{(t)}(\mu) \triangleq \mu - \frac{\eta}{pqn} \left(\sum_{i \in \mathcal{I}^{(t)}} \sum_{\xi \in \mathcal{B}_i^{(t)}} \text{clip}_c(\nabla l(\mu; \xi)) + \frac{\sqrt{\beta^{(t)}}}{\gamma^{(t)}} \zeta^{(t)} \right),$$

as the wireless noisy mechanism based on the dataset $\mathcal{D}$ at iteration t , and we can similarly define $\psi^{(t)'}(\cdot)$ based on its adjacent dataset $\mathcal{D}'$. If Assumption 1 holds, then for any random variables μ and ν , and for any $\beta^{(t)} \in (0, 1]$, we have

$$\mathcal{D}_{\alpha}^{((1+\eta L)z)}(\mathbb{P}_{\psi^{(t)}(\mu)} \parallel \mathbb{P}_{\psi^{(t)'(\nu)}}) \leq \mathcal{D}_{\alpha}^{(z)}(\mathbb{P}_{\mu} \parallel \mathbb{P}_{\nu}) + \frac{2\alpha pq c^2 (\gamma^{(t)})^2}{\beta^{(t)} \sigma^2}.$$

3. Control of Shift Amount

Lemma 2.6. (Shift-reduction (Feldman et al. 2018)). Let μ, ν be two random variables. Then, for any $a \geq 0$ and $z \geq 0$, we have

$$\mathcal{D}_{\alpha}^{(z)}(\mathbb{P}_{\mu} * \mathbb{P}_{\zeta} \parallel \mathbb{P}_{\nu} * \mathbb{P}_{\zeta}) \leq \mathcal{D}_{\alpha}^{(z+a)}(\mathbb{P}_{\mu} \parallel \mathbb{P}_{\nu}) + \frac{\alpha a^2}{2\sigma^2}, \quad (5)$$

where ζ is a multi-variate Gaussian random variable with zero mean and covariance given by $\sigma^2 \mathbf{I}_d$, denoted by $\zeta \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_d)$; and $\mathbb{P}_{\mu} * \mathbb{P}_{\zeta}$ denotes the distribution of the sum $\mu + \zeta$ with μ and ζ being drawn independently.

III. Privacy Analysis for DPWFL

4. Privacy Guarantee for DPWFL

Theorem 3.1 ((α, ϵ)-RDP Guarantee for DPWFL) Under Assumptions 1 and 2, for any $\alpha > 1$, DPWFL satisfies (α, ϵ)-RDP, where

$$\epsilon = \frac{2\alpha pq c^2}{\sigma^2} \min \left\{ \sum_{t=0}^{T-1} (\gamma^{(t)})^2, \Phi \right\}, \quad (4)$$

with $\Phi = (\gamma^{(T-1)})^2 + \frac{(1+\eta L)\sqrt{pq}Dn\gamma^{(T-1)}}{2\eta c}$.

- This result shows that, even in the wireless FL setting, the DPWFL can achieve a **converged privacy loss** with bounded domain assumption. This stands in sharp contrast to prior analyses of wireless FL [1], [2], where privacy loss goes to infinity with the number of iterations.
- Moreover, the presence of p and q in the bound explicitly quantifies the **privacy amplification by device selection and mini-batch sampling**, respectively.

Proof Sketch:

- Recursive Hypothesis
- Auxiliary variables

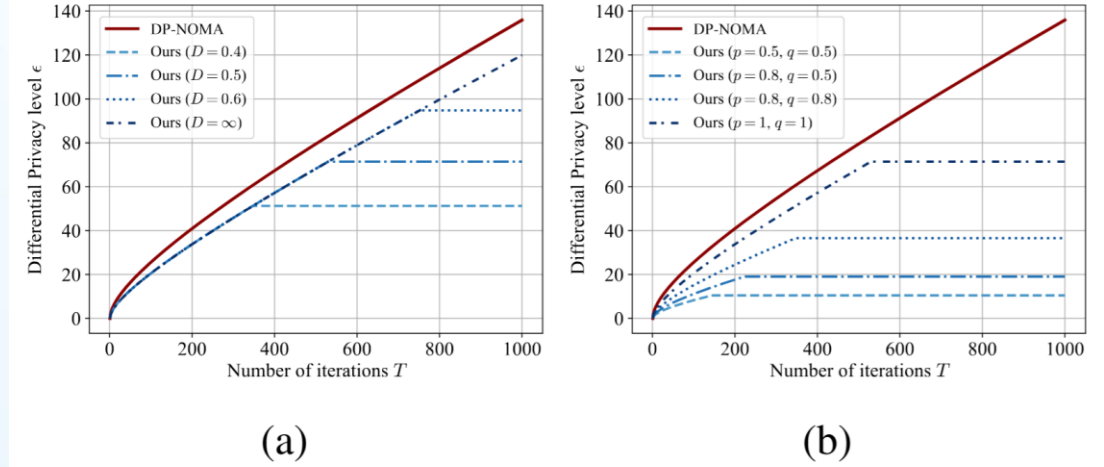


Fig. 1. The evolution of the privacy level during DPWFL with different settings: (a) domain diameter, D , (b) device sampling rate, p , and mini-batch sampling rate, q .

III. Convergence Analysis for DPWFL

5. Convergence Analysis for DPWFL

Theorem 3.2 (Convergence Analysis for DPWFL) Under Assumptions 1, 3, and 4, the DPWFL algorithm running for T global rounds with step-size $\eta = \mathcal{O}(1/L)$ satisfies

$$\begin{aligned} \min_{t \in [1, T]} \mathbb{E} \left[\|\nabla f(\boldsymbol{\theta}^{(t)})\| \right] &\leq \underbrace{\mathcal{O} \left(\frac{f(\boldsymbol{\theta}^{(0)}) - f^*}{\eta c T} + \sqrt{\frac{f(\boldsymbol{\theta}^{(0)}) - f^*}{\eta T}} \right)}_{\text{Initialization error } C_1} \\ &+ \underbrace{\sqrt{\frac{\eta L(\sigma_l^2 + \sigma_g^2)}{pqn}} + \min \left\{ \eta L \sqrt{\sigma_l^2 + \sigma_g^2}, c \right\}}_{\text{Stochastic noise \& dissimilarity \& clipping error } C_2} \\ &+ \underbrace{\min \left\{ \sigma_l + \sigma_g, \frac{\sigma_l^2 + \sigma_g^2}{c} \right\}}_{\text{Unavoidable clipping bias } C_3} \\ &+ \underbrace{\frac{d\eta L \sigma^2}{pqnc} \frac{1}{T} \sum_{t=0}^{T-1} \frac{1}{(\gamma^{(t)})^2} + \sqrt{\frac{d\eta L}{pqn} \frac{1}{T} \sum_{t=0}^{T-1} \frac{\sigma}{\gamma^{(t)}}}}_{\text{Estimation \& channel noise error } C_4}. \end{aligned}$$

Proof Sketch:
Discussion for
Different Cases

$$\begin{aligned} \mathcal{J}_{c+} &:= \{t; \|\nabla f(\bar{\mathbf{x}}_t)\| \geq c\}, \\ \mathcal{J}_{\frac{c}{2}+} &:= \{t; \frac{c}{2} \leq \|\nabla f(\bar{\mathbf{x}}_t)\| < c\}, \\ \mathcal{J}_{\frac{c}{2}-} &:= \{t; \|\nabla f(\bar{\mathbf{x}}_t)\| < \frac{c}{2}\}. \end{aligned}$$

When $c > \mathcal{O}(\sigma + \zeta)$

When $c < \mathcal{O}(\sigma + \zeta)$

- the first two terms (initialization error): capture optimization-related factors
- the third term: accounts for the bias introduced by stochastic gradient randomness, gradient dissimilarity, and clipping operation
- the fourth term: unavoidable bias from gradient clipping
- the last two terms (wireless communication error): arising from received signal estimation and channel noise

III. Privacy-Utility Trade-off for DPWFL

6. Privacy-Utility Trade-Off for DPWFL

Proposition 3.2 (Privacy-Utility Trade-off for DPWFL)

Under Assumptions 1-4, the DPWFL algorithm is (α, ε) -RDP, and satisfies

$$\begin{aligned} \min_{t \in [1, T]} \mathbb{E} \left[\|\nabla f(\boldsymbol{\theta}^{(t)})\| \right] &\leq \mathcal{O} \left(C_1 + C_2 + C_3 \right. \\ &\left. + \frac{\alpha d \eta L}{n \varepsilon} \frac{c}{T} \sum_{t=0}^{T-1} \frac{\Gamma}{(\gamma^{(t)})^2} + \sqrt{\frac{\alpha d \eta L}{n \varepsilon}} \frac{c}{T} \sum_{t=0}^{T-1} \frac{\sqrt{\Gamma}}{\gamma^{(t)}} \right). \end{aligned} \quad (6)$$

Equivalently, the DPWFL is (ϵ, δ) -DP, and its privacy-utility trade-off can be directly obtained by a similar analysis.

- Our analysis **rigorously incorporates the clipping effect** into the convergence analysis;
- The bound explicitly reveals the **inherent privacy-utility trade-off**, where tighter privacy inevitably degrades convergence;
- These results **can be further optimized and simplified** by carefully selecting the iteration number T and clipping threshold c .

Contents

- I. Introduction
- II. Related Works and Motivations
- III. The Proposed Method
- IV. Numerical Results

IV. Numerical Results

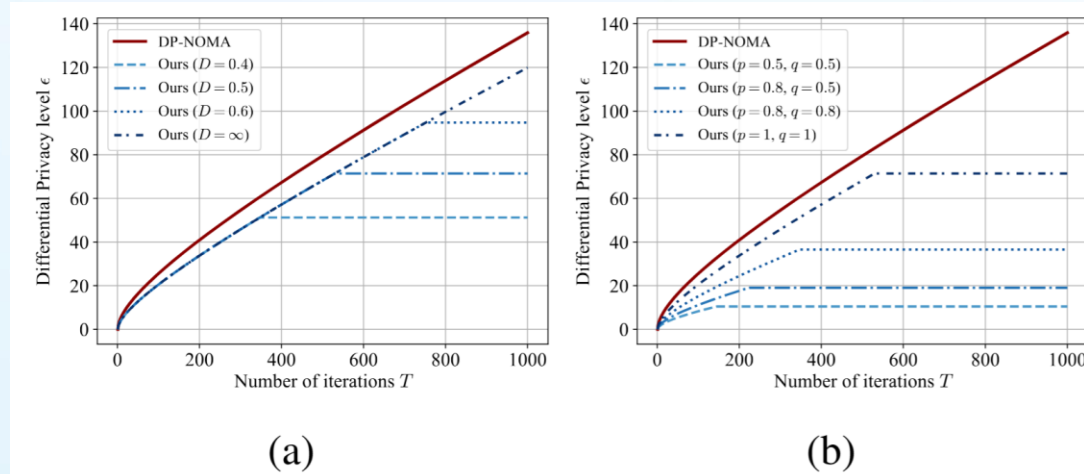


Fig. 1. The evolution of the privacy level during DPWFL with different settings: (a) domain diameter, D , (b) device sampling rate, p , and mini-batch sampling rate, q .

- Smaller domain diameters D yield a smaller converged privacy loss
- Smaller device selection rates p or mini-batch sampling rates q may lead to smaller ϵ
- Our privacy bound consistently outperforms DP-NOMA

The codes are available at <https://github.com/HauLiang/DPWFL>



2026 IEEE INTERNATIONAL CONFERENCE ON
ACOUSTICS, SPEECH, AND SIGNAL PROCESSING (ICASSP 2026)

May 04 - 08, 2026

Barcelona, Spain



Thank you!

Correspondence to: reikochen1085@gmail.com, haoliang@stu.xmu.edu.cn, xttu@xmu.edu.cn

Codes: <https://github.com/HauLiang/DPWFL>

MLSP-L31.4: Optimization and Privacy in Federated Learning

Fri, 8 May, 14:00 - 16:00

2026.ieeeicassp.org